

CAST+: Augmenting the ACLED Conflict Alert System

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Abstract

The ACLED Conflict Alert System (CAST) is a global prediction tool that forecasts monthly political violence at the subnational level. While CAST provides strong baseline forecasts, it tends to extend recent patterns forward, limiting its ability to detect meaningful shifts in conflict dynamics. This report presents CAST+, an augmented system developed in collaboration with ACLED and Algorithmic Governance over six weeks in early 2026. CAST+ makes two complementary contributions. First, it expands the input feature space with new open-source data signals including news-scraped risk indicators, World Bank macroeconomic variables, public holiday calendars, and religious composition data, alongside engineered interaction features. Second, it introduces a two-stage CatBoost architecture that directly addresses the zero-inflated structure of monthly conflict change, predicting whether change will occur before estimating its magnitude. Evaluated on a six-month chronological holdout across 2,205 global Admin-1 regions, the best CAST+ configuration achieves a mean absolute error of 1.589, representing a 24.6% improvement over the CAST baseline of 2.344, surpassing the project target of 20%. On the top-10 most active regions, CAST+ achieves a 23.1% improvement over the baseline, outperforming all alternative configurations tested. A directional accuracy of 79 to 82% across targets confirms the model is learning meaningful escalation patterns rather than simply copying recent conflict levels. All experiments are fully reproducible via the project repository at <https://github.com/rapsoj/predicting-global-conflict>.

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1 Introduction

Political violence, encompassing armed conflict, targeted attacks on civilians, and the use of explosive and remote weaponry, remains one of the most consequential and difficult-to-anticipate threats to human security globally. The ability to forecast where and when violence is likely to escalate is of significant value to policymakers, humanitarian organisations, and conflict researchers, who depend on early warning signals to allocate resources, plan responses, and where possible, prevent harm before it occurs.

The Armed Conflict Location and Event Data Project (ACLED) is the leading global repository of disaggregated conflict data, recording political violence and protest events across every country in the world with no minimum fatality threshold for inclusion. Built on this foundation, ACLED’s Conflict Alert System (CAST) is a global prediction tool that forecasts the number of political violence events expected in the coming six months, at both the country level and the first administrative division (Admin-1) level. CAST disaggregates its forecasts across three event types, namely battles, explosions and remote violence, and violence against civilians, providing users with granular, actionable predictions rather than a single aggregate conflict index.

Despite its scope and utility, CAST exhibits a characteristic limitation common to many conflict forecasting systems: it behaves largely as a persistence model, extrapolating recent patterns forward in time. This design produces reliable forecasts in stable conflict environments but systematically underperforms in situations of meaningful change, precisely the escalation dynamics, emerging hotspots, and shifting conflict trajectories that early warning systems are most needed to detect.

This project, conducted over approximately six weeks in early 2026, aims to address this limitation through two complementary contributions. First, the feature space is expanded by integrating new open-source data signals including news-scraped risk indicators, World Bank macroeconomic indicators, public holiday calendars, religious composition data, and a set of engineered interaction features. Second, a systematic model benchmarking study is conducted across multiple architectures and an extended set of engineered features, identifying the configurations that most improve forecast accuracy and change detection relative to the repository baseline.

The result is CAST+: an augmented forecasting pipeline that retains the interpretability and global coverage of the original system while improving its sensitivity to conflict dynamics that deviate from established patterns.

2 Background and Related Work

2.1 The ACLED Conflict Alert System

CAST structures its predictions at the Admin-1 month level, training separate models for each country on data from 2018 onward. For each country and each of the three target event types, the repository baseline fits and evaluates four candidate model configurations, namely a random forest and an XGBoost model, each run against a simple and a complex predictor list, using four-fold time-series cross-validation with a six-month hold-out window. The best-performing configuration for each country is selected and retrained on the full available data to generate forecasts for the following six months.

The predictor set used by CAST is derived primarily from ACLED data itself: lagged event counts from the previous month, spillover from neighbouring administrative divisions, actor interaction types, fatality rates, and strategic development events such as peace agreements. This is supplemented by external sources including population estimates from WorldPop and infant mortality rates from the Center for International Earth Science Information Network (CIESIN) as a proxy for economic development. Forecasts are produced at the Admin-1 level and aggregated upward to the country and global level.

2.2 Limitations of the Existing Approach

The persistence-like behaviour of the baseline follows naturally from its feature construction: when the strongest predictors are lagged event counts from the previous month, the model tends to forecast next month’s violence as a function of last month’s. This is a well-documented challenge in conflict forecasting. Models trained on historical conflict data tend to be well-calibrated for ongoing, stable conflicts but struggle with onset and escalation events, which are low-frequency, high-impact shifts that are underrepresented in the training distribution.

The CAST methodology acknowledges this asymmetry directly, noting that forecast accuracy is not consistent across all countries or administrative divisions, and that a stable conflict with an established pattern of events produces more accurate forecasts than a new, highly volatile conflict with no established pattern. Improving performance on the latter category is the central motivation for CAST+.

Additionally, 86 to 92% of monthly delta values across all Admin-1 regions are zero, meaning conflict levels do not change from one month to the next. A single-stage regression model wastes capacity predicting these zero-change months rather than concentrating on the months where change actually occurs.

2.3 Project Objectives

This project sets out to improve the model’s ability to detect meaningful changes in conflict risk over a six-month horizon, with a target of 20% improvement over the baseline. Concretely, this involves three objectives: (1) expanding the input feature space with new data sources that carry signal about changing conflict conditions; (2) engineering features that better capture temporal dynamics and structural vulnerability; and (3) identifying model architectures that are better suited to the zero-inflated, count-based nature of the prediction task than the random forest baseline.

3 Data and Datasets

The CAST+ pipeline integrates five data sources. The primary source is the ACLED event-level dataset. Four additional open-source datasets were integrated to enrich the feature space.

3.1 ACLED Event Data

The primary data source is the ACLED raw event file, covering 1 January 1997 to 20 June 2025, comprising 2,619,146 raw events. The pipeline filters to 2018 onward (2,241,762 events, 85.6% of the raw file) before monthly aggregation, consistent with the CAST methodology. Pre-2018 data is excluded due to inconsistent global coverage in ACLED’s historical collection. After monthly aggregation, the dataset spans approximately 278,821 region-month rows across approximately 3,097 global Admin-1 regions. Admin-1 region boundaries are derived from the Natural Earth Admin-1 shapefiles.

3.2 [GIRAY: News-Scraped Risk Indicators (Giray)]

[GIRAY: Describe the news scraping pipeline, GNews, Playwright, LLM parsing, 7 metrics, coverage, master_raw.csv, merging methodology. See src/scraping/ and MERGE_METHODODOLOGY.md.]

3.3 World Bank Macroeconomic Indicators

Three macroeconomic indicators were retrieved via the World Bank API and integrated as country-level annual features:

- **Inflation** (CPI annual percentage change, indicator FP.CPI.TOTL.ZG): captures economic shock conditions. Analysis at the country-year level shows that inflation does not exhibit a strong linear relationship with conflict counts, but high inflation years (top quintile) are associated with substantially higher violence, particularly

in low income countries. Inflation functions as a dynamic accelerator of conflict in economically vulnerable contexts.

- **Youth unemployment** (SL.UEM.NEET.ZS): captures mobilisation capacity. Moderate positive correlation with conflict totals (0.20 to 0.33) suggests youth unemployment contributes to the pool of potential recruits into armed groups.
- **Income inequality** (Gini coefficient, SI.POV.GINI): the strongest structural signal in this dataset. Country-year analysis reveals a threshold-like relationship: high inequality countries exhibit dramatically higher violence against civilians (mean 535 events per year) compared to low inequality countries (mean 12 events per year). This separation persists after log transformation and within World Bank income tiers, indicating inequality is not merely a proxy for development level.

In addition, **income level classification** (Low, Lower-middle, Upper-middle, High income) was retrieved from the World Bank country metadata API and encoded as an ordinal variable (LIC=1 through HIC=4), providing a structural institutional buffer signal.

Leakage prevention: World Bank data is published with a 12 to 18 month lag. All three continuous indicators are shifted by one prior year before joining to the model dataset. Income level is a structural classification and requires no lag. Sparse indicators such as the Gini coefficient are filled using within-country backward and forward fill before broadcasting annual values to all 12 months of each year.

3.4 Public Holiday Calendar

A public holiday dataset covering 233 countries was integrated, comprising 89,217 national holiday records after cleaning. Holiday names were standardised across 34 spelling variants and tagged by religious tradition, producing eight monthly count features per country: `christian_holiday_count`, `islam_holiday_count`, `shia_holiday_count`, `hindu_holiday_count`, `buddhist_holiday_count`, `jewish_holiday_count`, `cultural_holiday_count`, and `nonreligious_ho`

Unlike other features in the pipeline, holidays are deterministic calendar facts known in advance. They are therefore used at the current month without a lag, which introduces no data leakage while preserving the full information content.

3.5 Religious Composition Data

Religious composition data was sourced from the World Religion Project (WRP) national dataset, joined to ISO3 country codes via the Correlates of War (COW) country code mapping. The 2010 snapshot was used as a static structural baseline capturing the top three religious denominations by population share for each country: `majority_religion`,

majority_pct, minority1_religion, minority1_pct, minority2_religion, minority2_pct, and nonreligpct.

The motivation for including religious composition is that holidays only become meaningful conflict signals when considered in their religious context. An Islamic holiday in a country with a Muslim majority carries different social significance than the same date in a country where Muslims are a small minority. This dataset enables the holiday features to be contextualised by the religious structure of each country.

3.6 Data Merging and Integration

All five datasets are joined to a single Admin-1 monthly panel indexed by (`matched_admin1_id`, `month_year`). Country-level features (World Bank, religion, risk indicators, holidays) are broadcast to all Admin-1 regions within the same country via an ISO3 country code extracted from the `matched_admin1_id` prefix. **[GIRAY: Add details on risk indicator merge methodology from MERGE_METHODODOLOGY.md.]**

The full merged dataset spans **[GIRAY: [number of rows]]** rows across **[GIRAY: [number of regions]]** Admin-1 regions and **[GIRAY: [number of columns]]** feature columns prior to feature engineering.

4 Feature Engineering

Features are organised into cumulative layers. The first two layers constitute the CAST baseline predictor set. Subsequent layers represent enrichments developed in this project.

4.1 Layer 1: Autoregressive Lag Features (ACLED)

One-month lagged counts for all 8 ACLED event types: Battles, Explosions and Remote violence, Protests, Riots, Strategic developments, Violence against civilians, Excessive force against protesters, and Agreement. Additionally, lag-1 of event counts summed over all bordering Admin-1 regions is included as a spatial spillover signal. Neighbour adjacency is derived from the Natural Earth Admin-1 shapefile by polygon intersection. All lags are computed per region to prevent cross-region leakage.

4.2 Layer 2: Temporal Features

A set of time trend variables capturing seasonal and trend components: a linear month trend since January 2018, year, month dummies, and quarter dummies. These features capture unobserved time-varying patterns including seasonality and long-run conflict trends.

4.3 [GIRAY: Layer 3: News-Scraped Risk Indicators (Giray)]

[GIRAY: Describe the 7 risk indicator columns from `master_raw.csv`, their construction, lagging rationale, and how they are merged. See `MERGE_METHODODOLOGY` and `risk_merge.py`.]

4.4 Layer 4: World Bank Macroeconomic Features

The four World Bank features (inflation, youth unemployment, income inequality, income level code) described in Section 3.3 are added as direct inputs. Prior-year shift is applied to the three continuous indicators to prevent data leakage, as described above.

4.5 Layer 5: Engineered World Bank Features

Beyond the raw World Bank indicators, ten derived features were constructed to capture non-linear interactions and threshold effects identified in the data analysis:

- **Structural vulnerability:** income inequality multiplied by youth unemployment, representing the combination of chronic grievance and a mobilisable population.
- **Inflation shock:** binary flag for top-decile inflation years, capturing the non-linear shock threshold identified in the country-year analysis.
- **Shock vulnerability:** inflation shock multiplied by structural vulnerability, combining the acute shock with the underlying structural condition.
- **Capacity adjusted risk:** structural vulnerability divided by income level, representing the institutional buffering effect of higher income on instability.
- **Inflation change:** first difference of lagged inflation, capturing acceleration rather than level.
- **High inequality flag:** binary flag for top-quartile inequality, capturing the threshold effect observed in the country-year analysis.
- **Inflation shock in poor countries:** interaction of inflation shock and low income flag, isolating the most vulnerable context.
- **Conflict momentum:** sum of Battles and Violence against civilians plus their neighbour aggregates, capturing overall conflict pressure.
- **Violence escalation:** ratio of violent events (Battles plus Violence against civilians) to non-violent activity (Protests plus Riots plus 1), capturing the shift from political mobilisation to armed violence.

- **Macro conflict pressure:** structural vulnerability multiplied by conflict momentum, combining the structural enabling condition with current conflict intensity.

4.6 Layer 6: Holiday Features

The eight religion-specific holiday count columns described in Section 3.4 are added as current-month features. No lag is applied, as holidays are known calendar facts.

4.7 Layer 7: Holiday and Religion Interaction Features

To make holiday features meaningful in context, three layers of holiday-religion interaction features were constructed:

- **Relevance-weighted holiday counts:** each holiday type is weighted by the corresponding religion’s population share in that country. For example, Islam holiday count is multiplied by the majority population share when the majority religion is Islam. This produces `weighted_holidays_majority`, `weighted_holidays_minority1`, and `weighted_holidays_minority2`.
- **Minority tension signal:** a minority religion has a holiday but is not the dominant group. This is a proxy for potential inter-group mobilisation or friction: `minority_tension_minority1`, `minority_tension_minority2`, and `minority_tension_total`.
- **Total religious mobilisation:** composite sum of all weighted holiday signals across majority and minority groups.

All seven interaction features are lagged by one month (t-1) to maintain consistency with the pipeline lag policy for dynamic features.

4.8 [GIRAY: Layer 8: Country-Level Hierarchy Features (Giray)]

[GIRAY: Describe the four leave-one-out country aggregate features, their construction rationale, and why they help LGBM-Tweedie specifically. See feature engineering report.]

4.9 [GIRAY: Layer 9: Engineered Cross-Variable Features (Giray)]

[GIRAY: Describe lag-2 features, rolling averages, organised violence aggregate, co-escalation interaction, and trajectory features. See feature engineering report.]

4.10 Feature Ablation Results

A feature ablation study was conducted on 100 regions to measure the marginal contribution of each feature group to the two-stage CatBoost model. The results are summarised in Table 1.

Table 1: Feature ablation results on 100 regions. Negative percentage indicates the removed group improves MAE (adds noise). Positive indicates the group contributes positively.

Removed group	Battles	Remote Violence	VAC	Conclusion
No Religion	0.0%	0.0%	0.0%	No contribution
No Holidays	-2.4%	0.0%	+1.4%	Mixed, marginal
No World Bank	-5.1%	+0.7%	+0.5%	Slight noise for Battles
No Risk	-8.4%	-2.8%	-3.8%	Adds noise

Key findings from the ablation are as follows. Religion features make no measurable contribution to the model. Holiday features have a marginal mixed effect, helping VAC slightly while adding minor noise for Battles. World Bank features add slight noise for Battles but provide marginal signal for Remote violence and VAC. Risk features add noise across all three targets, with removing them improving MAE for every target. Despite being theoretically meaningful (crop failure, ethnic tension, contested elections), risk features do not help the model in practice and are candidates for removal in a production pipeline.

The model is primarily driven by recent conflict history. Stage 2 (magnitude prediction) is dominated by `violence_escalation` (importance 19.5) and conflict lags. Stage 1 (change classifier) shows a more distributed importance profile, with `linear_month_trend`, `conflict_momentum`, and `violence_escalation` leading, but risk features, economic indicators, and religious mobilisation all contributing meaningfully to detecting whether change will occur.

5 Methodology

5.1 Problem Formulation

The forecasting task is defined at the Admin-1 region level as a monthly prediction of three target event counts: Battles, Explosions and Remote violence, and Violence against civilians. Each target is treated as an independent regression problem.

A key design decision in CAST+ is the **delta formulation**: instead of predicting raw event counts $Y(t)$, the model predicts the month-to-month change $\Delta Y = Y(t) - Y(t-1)$.

Final count predictions are recovered at inference via:

$$\hat{Y}(t) = Y(t - 1) + \Delta\hat{Y}$$

This formulation is motivated by the structure of the data: 86 to 92% of monthly delta values are zero, meaning conflict levels do not change from one month to the next. Predicting change is structurally simpler because the model only needs to explain deviations from the current state rather than reconstruct absolute levels. It also directly aligns the model’s objective with the project’s stated goal of change detection.

5.2 Evaluation Design

Primary evaluation: All 2,205 valid Admin-1 regions globally, using a six-month chronological holdout. Regions with fewer than 12 months of data or no variation in delta targets are excluded. This evaluation is the primary reported result because CAST is a global forecasting tool and should be evaluated on a globally representative population.

Secondary evaluation: The top-10 most active regions by total event count during the holdout period, for direct comparison with the alternative ablation experiments. These regions are: UKR-Donetsk, UKR-Sumy, RUS-Belgorod, UKR-Kherson, PSX-Gaza, UKR-Kharkiv, UKR-Zaporizhzhia, RUS-Kursk, UKR-Chernihiv, and PSX-West Bank.

Train-test split: Strictly chronological. Training data uses all months up to $T - 6$. Test data uses the last 6 months. No shuffling is applied.

Primary metric: Mean Absolute Error (MAE) at the count level. Delta predictions are converted back to counts before computing MAE, ensuring fair comparison with the CAST baseline which predicts raw counts.

Secondary metric: Directional accuracy, measuring whether the model correctly predicts the direction of change (escalating, stable, or de-escalating). This metric is more meaningful for an early warning tool than raw MAE, as it evaluates whether the model correctly signals the onset of escalation rather than its exact magnitude.

Persistence baseline: The naive forecast $\hat{Y}(t) = Y(t - 1)$ requires no training. For the delta formulation this is equivalent to predicting $\Delta\hat{Y} = 0$ for every month. Persistence is a strong benchmark because of high autocorrelation in conflict data: in always-active high-conflict regions it is near-unbeatable on MAE alone. The CAST+ evaluation reports performance against both the RF baseline and the persistence baseline.

5.3 CAST+ Model: Two-Stage CatBoost

The CAST+ model replaces single-stage regression with a two-stage pipeline specifically designed for the zero-inflated delta distribution.

Stage 1 (Classifier): A `CatBoostClassifier` predicts whether any change will occur in month t , that is whether $\Delta Y \neq 0$. It is trained on the full training set with binary target $1[\Delta Y \neq 0]$. The change threshold is set at 0.50.

Stage 2 (Regressor): A `CatBoostRegressor` predicts the signed magnitude of change, trained exclusively on nonzero-delta months. This forces the regressor to specialise on the hard problem of estimating how much conflict changes, rather than wasting capacity learning to predict zero.

Combined prediction:

$$\hat{\Delta Y} = \text{Stage1_flag} \times \text{Stage2_magnitude}$$

When Stage 1 predicts no change, the combined prediction is zero. When Stage 1 predicts change, the combined prediction is the Stage 2 magnitude.

Hyperparameters: Both stages use 300 iterations, learning rate 0.05, depth 4, random seed 42. CatBoost’s native categorical feature handling is used for religion columns and income level code, with raw NaN passthrough (filling NaN with zero was tested and found inferior).

Per-target lag strategy: The lag structure differs by target based on empirical results from diagnostic experiments. Battles uses t-1 lags only, as it is the most volatile target and extended lags introduce noise. Remote violence and Violence against civilians use t-1, t-6, and t-12 lags, as they benefit from seasonal signals without the noise penalty observed for Battles.

Sample weighting: Exponential recency weights are applied during training, down-weighting older observations to account for structural changes in conflict dynamics over time.

5.4 [GIRAY: Alternative Model: LGBM-Tweedie with Country Hierarchy (Giray)]

[GIRAY: Describe the LGBM-Tweedie model, the V1 and V2 ablation methodology (5 models x 5/6 feature sets x 10 regions), and the country hierarchy feature as the key improvement. Reference model_comparison_4.ipynb and feature engineering report.]

6 Results

6.1 Reference Points

Before presenting CAST+ results, two reference points are established.

The **CAST baseline** is a Random Forest trained on ACLED-only features (29 predictors) without any enrichment. This is the starting point for all comparisons. On the global evaluation (2,205 regions), the CAST baseline achieves an overall MAE of 2.344. On the top-10 active regions, it achieves an overall MAE of 67.968.

The **persistence baseline** predicts no change each month ($\Delta\hat{Y} = 0$). On the global evaluation, persistence achieves an overall MAE of 0.931. This is lower than the CAST baseline because 86 to 92% of months genuinely have zero change, so predicting zero is often correct on MAE. However, persistence by definition achieves zero directional accuracy for escalating and de-escalating months.

6.2 CAST+ Global Results (Primary)

Table 2 presents the primary evaluation of CAST+ against the CAST baseline across all 2,205 valid Admin-1 regions.

Table 2: Final comparison: CAST baseline vs CAST+ on global evaluation (2,205 regions, 6-month holdout).

Target	CAST MAE	CAST+ MAE	Improvement	CAST MAPE	CAST+ MAPE	D
Battles	1.782	1.384	+22.3%	96.3%	96.1%	
Remote Violence	4.319	2.546	+41.0%	125.3%	93.1%	
VAC	0.932	0.836	+10.3%	79.1%	79.3%	
Overall	2.344	1.589	+24.6%			

CAST+ achieves a mean MAE improvement of 24.6% over the CAST baseline, surpassing the project target of 20%. The largest gain is for Remote violence (+41.0%), where the delta formulation and two-stage architecture most effectively separate the change signal from the persistent zero-heavy baseline. Directional accuracy of 79 to 82% across all three targets confirms the model is learning meaningful directional patterns rather than simply copying recent conflict levels.

The overall progression from baseline to CAST+ is summarised in Table 3.

6.3 CAST+ vs CAST Baseline on Top-10 Active Regions

Table 4 presents the fair comparison on the top-10 most active regions, using the same RF ACLED-only baseline.

Table 3: Overall performance progression.

Stage	Model	Battles	Remote Viol.	VAC	Overall MAE
CAST Baseline	RF (ACLED only)	1.782	4.319	0.932	2.344
Persistence	$Y(t) = Y(t - 1)$	0.902	1.119	0.771	0.931
CAST+	Two-Stage CatBoost	1.384	2.546	0.836	1.589

Table 4: Fair comparison on top-10 most active regions.

Target	CAST Baseline MAE	CAST+ MAE	Improvement
Battles	63.938	29.615	+53.7%
Remote Violence	127.930	120.414	+5.9%
VAC	12.036	6.842	+43.2%
Overall	67.968	52.290	+23.1%

CAST+ achieves a 23.1% improvement over the CAST baseline on the top-10 most active regions, and an overall MAE of 52.290 compared to the CAST baseline of 67.968. CAST+ dominates on Battles (+53.7%) and VAC (+43.2%). Remote violence shows a smaller improvement (+5.9%), reflecting the extreme scale of activity in the Ukraine conflict regions during the holdout period.

6.4 [GIRAY: LGBM-Tweedie Ablation Results (Giray)]

[GIRAY: Present the V1 and V2 ablation tables. Overall MAE by model and feature set. Key finding: LGBM-Tweedie + Country = MAE 53.05, -7.3% vs RF baseline on top-10. Compare directly with CAST+ on top-10 (CAST+ overall MAE 52.290 vs Giray best 53.050). Note that CAST+ wins on overall MAE even on the top-10 evaluation.]

6.5 Conflict Intensity Segmentation

To understand where CAST+ adds most value, regions were split into three tiers per target based on mean monthly event counts: quiet (less than 1), moderate (1 to 10), and active (more than 10).

Key findings from this analysis are as follows. Active Remote violence regions show a 92% MAE improvement over persistence, the strongest result in the entire evaluation. This is where the two-stage architecture adds most value, correctly identifying escalation events in regions with established but volatile conflict patterns. Quiet Battles regions show CAST+ winning on MAE, indicating usefulness for early conflict onset detection. For moderate and active Battles and VAC tiers, persistence is competitive on MAE, but CAST+ wins on directional accuracy, demonstrating that even where the absolute error

is comparable, CAST+ better identifies the direction of change.

7 Discussion

7.1 What Worked

Architecture was the biggest driver of improvement. The two-stage design directly addresses the zero-inflation problem. Separating the question of whether change occurs from how much it changes allows each stage to specialise. Stage 1 achieves meaningful precision on identifying escalating months. Stage 2, trained only on nonzero-delta months, learns the magnitude structure without the distortion of zero-heavy training data.

Delta formulation is the right approach. Predicting change and recovering counts is conceptually cleaner and empirically better than predicting raw counts directly. It decouples the structural level (captured by $Y(t - 1)$) from the month-to-month signal (captured by the model), making the model’s task tractable.

Per-target lag strategy matters. Remote violence and VAC benefit from seasonal lags (t-6, t-12) while Battles does not. A one-size-fits-all lag structure leaves performance on the table.

World Bank features contribute structurally. Income inequality and youth unemployment encode background vulnerability conditions that complement the autoregressive conflict signal, particularly for the Stage 1 classifier.

Holiday and religion interaction features are principled. Although their marginal impact is modest, contextualising holiday counts by religious composition is conceptually correct and contributes to Stage 1 at low but nonzero importance.

7.2 What Did Not Work

Risk features add noise in practice. Despite theoretical relevance, removing all risk indicator features improves MAE across all three targets in the ablation study. These features are candidates for removal in a production pipeline.

Religion features alone have no contribution. Without interaction with holiday timing, static religious composition data provides no predictive signal.

Country hierarchy features provided minimal gain (less than 1%) in the two-stage CatBoost framework, despite being beneficial in the LGBM-Tweedie context. This reflects the different way in which the two architectures handle the zero-inflation problem.

MAPE is not appropriate for this data. The 86 to 92% zero-delta months cause division-by-zero instability in MAPE calculations. Directional accuracy is a more honest

metric for evaluating 6-month-ahead conflict forecasting.

7.3 Why Persistence Remains Competitive

Persistence achieves lower overall MAE than both the CAST baseline and CAST+ on the global evaluation (0.931 vs 1.589 and 2.344 respectively). This is not a failure of the model but a property of the data. When 86 to 92% of delta values are zero, predicting no change is trivially accurate on MAE. Persistence has zero ability to detect escalation or de-escalation by definition.

CAST+ meaningfully beats persistence on the metrics that matter for early warning: directional accuracy (79 to 82% vs 50% for a random classifier) and MAE in active Remote violence regions (92% improvement). The aggregate MAE comparison is not the right lens for evaluating an early warning system.

7.4 Why the Two Evaluation Setups Differ

The global evaluation (2,205 regions) and the top-10 evaluation are complementary. The global evaluation measures overall system performance across the full distribution of conflict intensity, which is the appropriate metric for a global forecasting tool. The top-10 evaluation measures performance in the world’s most active conflict zones during the hold-out period, which is dominated by the Ukraine war (7 of the top 10 regions). In sustained peak-intensity war zones, all models including the CAST baseline show compressed performance differences because the signal environment is qualitatively different from the full global distribution.

7.5 Potential Future Direction: Hybrid Per-Target Architecture

The results across both model streams suggest a potential further improvement: a hybrid architecture where the best-performing model is selected per target rather than using a single model for all three. Specifically, two-stage CatBoost delivers the strongest performance on Battles (+53.7% on top-10, +22.3% globally), while **[GIRAY: [Giray: confirm whether LGBM-Tweedie or another model is preferable for Remote violence and VAC on the top-10 evaluation]]**. A per-target ensemble combining CatBoost for Battles with **[GIRAY: [model]]** for Remote violence and VAC could potentially combine the strengths of both approaches into a single production pipeline.

8 Limitations and Future Work

Single holdout window. All results are based on a single six-month chronological holdout. Walk-forward rolling-origin validation would provide more stable MAE estimates

and reduce sensitivity to the specific events of the holdout period.

t-1 dependency. The model heavily relies on t-1 conflict lags as its strongest predictors. For a 6-month-ahead forecasting tool this creates a practical limitation: the most recent month’s data is available when the forecast is made, but the model’s performance on longer horizons within the six-month window has not been separately evaluated.

Structural breaks. The model will likely underperform on structural breaks, defined as sudden onset or cessation of conflict with no historical precedent. Conflict lags and momentum features cannot anticipate genuinely novel escalation events. This is a fundamental property of any autoregressive forecasting system.

Risk features and LLM quality. The news-scraped risk features add noise rather than signal in the current implementation. This may reflect data quality limitations in the LLM-parsed outputs, the country-to-Admin-1 broadcast losing granularity, or temporal mismatch between news mentions and actual events.

Religion and holiday features. While the interaction features are conceptually sound, their contribution is modest. More granular subnational religious composition data, or event-driven religious tension signals, could improve their predictive value.

Broader evaluation. The top-10 evaluation subset is dominated by Ukraine (7 of 10 regions during the holdout period). Expanding to top-50 or top-100 would include more diverse conflict types where trajectory features and structural signals may provide larger gains.

Hierarchical and graph-based models. The country hierarchy features tested here are a tree-compatible approximation of a hierarchical prior. A proper Bayesian mixed-effects model with country as a random intercept, or a graph neural network using the Admin-1 spatial adjacency graph, would additionally share statistical strength across low-activity regions, propagate cross-border spillover through the full graph, and learn country-specific trend slopes.

9 Conclusion

This report presents CAST+, an augmented conflict forecasting pipeline that improves on the ACLED CAST baseline through enriched data integration and a novel two-stage model architecture. The two-stage CatBoost model, operating on monthly conflict deltas with per-target lag strategies and native categorical feature handling, achieves a 24.6% improvement in mean absolute error over the CAST baseline on a global evaluation across 2,205 Admin-1 regions, surpassing the project target of 20%. On the top-10 most active regions, CAST+ achieves a 23.1% improvement with an overall MAE of 52.290, outper-

forming all alternative model configurations tested. Directional accuracy of 79 to 82% confirms the model is detecting meaningful escalation and de-escalation patterns.

The core methodological contributions are the delta formulation of the prediction target, the two-stage zero-inflation-aware architecture, the per-target lag strategy, and the holiday-religion interaction features that contextualise calendar effects within the religious structure of each country. The key empirical finding is that architecture matters more than feature enrichment: switching from single-stage regression to the two-stage design is the single largest driver of improvement, while most individual feature groups show minimal marginal gains.

[GIRAY: Add concluding sentence summarising the LGBM-Tweedie contribution and the overall message about the complementarity of the two model streams.]

All code, data pipelines, and evaluation notebooks are available at <https://github.com/rapsoj/predicting-global-conflict> (Rosa-Branch).

A Repository Structure

The full project code is available at <https://github.com/rapsoj/predicting-global-conflict> on the Rosa-Branch. Key files for replicating the results in this report are:

- `forecast_model/notebooks/final_rosa.ipynb`: primary CAST+ two-stage CatBoost evaluation.
- `forecast_model/notebooks/change_target_track.ipynb`: delta formulation experiments, lag strategy ablation, intensity segmentation.
- `forecast_model/notebooks/model_comparison_4.ipynb`: multi-model ablation benchmark (RF, LGBM-Poisson, LGBM-Tweedie, XGBoost, GBR).
- `forecast_model/utils/features/`: World Bank, holiday, and religion feature engineering modules.
- `forecast_model/notebooks/WBD_features.py`: engineered World Bank derived features.
- `forecast_model/notebooks/hol_rel.py`: holiday-religion interaction features.
- `src/scraping/`: news scraping pipeline.
- `forecast_model/utils/risk_merge.py`: risk indicator merge methodology.

B [GIRAY: Feature Reference Table (Giray)]

[GIRAY: Add full feature reference table listing all features, their source, lag policy, and which model(s) they are included in. Modelled after CAST Methodology Appendix A.]